

## SKOVDE, Sweden

WMO: 025350

Lat: 58.456N

Lon: 13.973E

Elev: 324

StdP: 14.52

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 3.2        | 9.0 | -0.5                          | 5.4 | 4.0  | 5.4 | 7.3 | 11.2 | 21.1                  | 39.2 | 19.4 | 38.4 | 2.3                   | 180  | 0.607 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.3                   | 80.5            | 63.3 | 77.0 | 61.9 | 73.5 | 61.0 | 66.5                | 74.7 | 64.8 | 72.8 | 63.0 | 70.3 | 7.6                  | 100  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 64.2                            | 90.9 | 68.9 | 62.3 | 85.0 | 67.5 | 60.5 | 79.5 | 66.0 | 31.3          | 74.3 | 29.9 | 72.3 | 28.6 | 69.7 | 76.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 19.8          | 17.6 | 15.8 | DB                         | -1.2 | 85.2               | 4.9 | 3.7                                                | -4.7 | 87.9       | -7.5  | 90.1       | -10.3 | 92.1       | -13.8 | 94.9 | (4) |
| (5)               |      |      | WB                         | -3.1 | 69.4               | 6.2 | 2.2                                                | -7.6 | 70.9       | -11.2 | 72.2       | -14.7 | 73.4       | -19.2 | 75.0 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 44.4        | 29.1  | 28.6  | 34.6  | 42.8  | 51.4  | 56.9  | 61.5  | 59.3  | 53.1  | 44.7  | 37.9  | 32.4  | (6)  |  |
|                                                                        | DBStd       | 13.42       | 9.35  | 9.00  | 7.52  | 6.82  | 7.20  | 5.31  | 4.70  | 4.49  | 5.42  | 6.45  | 7.28  | 9.36  | (7)  |  |
|                                                                        | HDD50       | 3160        | 649   | 598   | 477   | 231   | 71    | 6     | 0     | 1     | 31    | 187   | 364   | 545   | (8)  |  |
|                                                                        | HDD65       | 7537        | 1114  | 1018  | 941   | 666   | 424   | 249   | 131   | 184   | 359   | 629   | 812   | 1010  | (9)  |  |
|                                                                        | CDD50       | 1132        | 0     | 0     | 1     | 15    | 113   | 212   | 356   | 289   | 123   | 22    | 1     | 0     | (10) |  |
|                                                                        | CDD65       | 36          | 0     | 0     | 0     | 0     | 1     | 5     | 22    | 7     | 1     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH74       | 652         | 0     | 0     | 0     | 1     | 59    | 151   | 355   | 80    | 6     | 0     | 0     | 0     | (12) |  |
|                                                                        | CDH80       | 107         | 0     | 0     | 0     | 0     | 3     | 23    | 70    | 11    | 0     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 7.2         | 7.7   | 7.4   | 7.6   | 7.1   | 6.8   | 6.8   | 6.3   | 6.4   | 7.0   | 7.3   | 7.9   | 7.8   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 27.0        | 1.7   | 1.6   | 1.2   | 1.5   | 2.1   | 3.3   | 3.4   | 3.1   | 2.5   | 2.4   | 2.2   | 2.0   | (15) |  |
|                                                                        | PrecMax     | 31.9        | 3.7   | 3.0   | 2.2   | 4.1   | 3.5   | 6.1   | 6.8   | 7.0   | 5.7   | 4.9   | 4.3   | 3.7   | (16) |  |
|                                                                        | PrecMin     | 20.9        | 0.4   | 0.1   | 0.2   | 0.1   | 0.6   | 0.4   | 0.6   | 0.6   | 0.4   | 0.4   | 0.4   | 0.7   | (17) |  |
|                                                                        | PrecStd     | 3.0         | 0.9   | 0.7   | 0.5   | 1.1   | 0.7   | 1.5   | 1.7   | 1.6   | 1.6   | 1.0   | 1.1   | 0.9   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 46.7  | 48.5  | 58.9  | 71.2  | 79.1  | 82.8  | 85.9  | 80.5  | 73.4  | 64.1  | 55.0  | 51.8  | (19) |  |
|                                                                        |             | MCWB        | 44.0  | 42.3  | 48.1  | 55.7  | 60.7  | 62.1  | 64.4  | 64.4  | 62.9  | 57.7  | 52.0  | 49.0  | (20) |  |
|                                                                        | 2%          | DB          | 44.7  | 44.8  | 53.4  | 66.0  | 75.2  | 78.6  | 81.0  | 75.6  | 68.4  | 59.0  | 52.0  | 49.6  | (21) |  |
|                                                                        |             | MCWB        | 42.6  | 41.6  | 45.3  | 52.1  | 57.4  | 60.4  | 64.3  | 63.7  | 60.0  | 54.9  | 49.4  | 47.1  | (22) |  |
|                                                                        | 5%          | DB          | 42.8  | 42.8  | 48.6  | 61.2  | 71.6  | 73.8  | 78.5  | 71.9  | 65.8  | 56.9  | 50.0  | 46.5  | (23) |  |
|                                                                        |             | MCWB        | 40.6  | 40.4  | 42.5  | 50.0  | 57.7  | 60.2  | 63.3  | 61.7  | 58.4  | 53.1  | 48.0  | 44.5  | (24) |  |
|                                                                        | 10%         | DB          | 40.9  | 40.8  | 46.1  | 57.1  | 66.4  | 69.9  | 73.8  | 69.5  | 62.6  | 54.0  | 48.0  | 44.4  | (25) |  |
|                                                                        |             | MCWB        | 38.9  | 38.4  | 41.5  | 47.4  | 55.4  | 58.2  | 62.6  | 61.1  | 56.4  | 50.8  | 46.5  | 42.3  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 44.7  | 44.6  | 49.2  | 56.5  | 64.3  | 66.1  | 69.4  | 68.6  | 64.5  | 58.6  | 52.8  | 49.7  | (27) |  |
|                                                                        |             | MCDB        | 45.8  | 46.7  | 56.4  | 68.3  | 73.4  | 76.0  | 77.0  | 73.8  | 70.3  | 61.5  | 54.5  | 51.1  | (28) |  |
|                                                                        | 2%          | WB          | 42.5  | 42.1  | 46.2  | 53.8  | 61.4  | 63.3  | 67.1  | 65.7  | 61.3  | 55.6  | 50.2  | 46.8  | (29) |  |
|                                                                        |             | MCDB        | 43.7  | 43.8  | 51.1  | 63.9  | 70.2  | 72.3  | 76.0  | 71.6  | 66.7  | 58.5  | 51.5  | 48.6  | (30) |  |
|                                                                        | 5%          | WB          | 40.6  | 40.1  | 43.7  | 51.0  | 59.0  | 61.2  | 65.3  | 63.9  | 59.2  | 53.7  | 48.2  | 44.4  | (31) |  |
|                                                                        |             | MCDB        | 42.0  | 42.2  | 47.2  | 59.1  | 66.5  | 71.2  | 74.6  | 69.5  | 64.1  | 56.3  | 49.5  | 45.9  | (32) |  |
|                                                                        | 10%         | WB          | 38.6  | 38.0  | 41.5  | 48.7  | 56.6  | 59.2  | 63.6  | 62.3  | 57.5  | 51.8  | 46.0  | 41.9  | (33) |  |
|                                                                        |             | MCDB        | 40.1  | 39.9  | 44.6  | 56.0  | 65.0  | 68.2  | 72.3  | 67.6  | 61.5  | 54.4  | 47.3  | 43.4  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 9.9   | 11.1  | 14.6  | 21.3  | 21.4  | 20.4  | 19.3  | 18.0  | 16.4  | 13.2  | 8.9   | 9.1   | (35) |  |
|                                                                        |             | MCDBR       | 9.5   | 12.4  | 19.4  | 29.0  | 29.5  | 29.0  | 28.3  | 23.2  | 20.8  | 14.0  | 9.3   | 8.8   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 9.2   | 9.8   | 13.0  | 17.5  | 15.4  | 15.2  | 13.6  | 13.1  | 13.3  | 10.7  | 8.4   | 8.9   | (37) |  |
|                                                                        |             | MCWBR       | 9.4   | 11.5  | 16.8  | 25.3  | 23.0  | 24.3  | 23.7  | 19.4  | 18.6  | 13.1  | 9.3   | 8.8   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.275 | 0.311 | 0.331 | 0.370 | 0.367 | 0.356 | 0.378 | 0.371 | 0.342 | 0.331 | 0.299 | 0.259 | (40) |  |
|                                                                        |             | taud        | 2.240 | 2.310 | 2.369 | 2.331 | 2.398 | 2.448 | 2.405 | 2.423 | 2.494 | 2.444 | 2.313 | 2.173 | (41) |  |
|                                                                        | Ebn at Noon | Ebn at Noon | 179   | 228   | 259   | 266   | 275   | 279   | 269   | 263   | 254   | 220   | 168   | 142   | (42) |  |
|                                                                        |             | Edh at Noon | 15    | 22    | 28    | 34    | 34    | 33    | 33    | 30    | 24    | 19    | 14    | 11    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 110   | 297   | 779   | 1254  | 1638  | 1836  | 1640  | 1298  | 881   | 399   | 142   | 75    | (44) |  |
|                                                                        | RadStd      | RadStd      | 18    | 49    | 81    | 149   | 141   | 93    | 201   | 126   | 113   | 64    | 26    | 8     | (45) |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (47) | Regional (59 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page